

FORMULA 1

2026 SEASON

Understanding the Broadcast Graphics

A Beginner's Guide to Formula 1

Introduction

Formula 1's live broadcast uses a rich set of on-screen graphics to convey race information in real time. Understanding these graphics transforms the viewing experience from passive watching to informed analysis. This guide covers every major graphic element you'll see during a 2026 broadcast.

2026 Broadcast Note

Apple TV+ becomes a major broadcast partner for F1 starting in 2026, bringing updated graphics packages and presentation styles. Core information elements remain consistent, though visual design may evolve throughout the season.

The Timing Tower

The timing tower is the vertical leaderboard displayed on the left side of the screen. It is the single most important graphic for following the race.

Information Displayed

Element	Description	Example
Position	Current race position (P1, P2, etc.)	1
Driver Code	Three-letter abbreviation	VER, HAM, NOR
Team Color	Colored bar showing team identity	Red Bull = dark blue
Interval	Gap to car directly ahead	+1.234
Gap to Leader	Total gap to the race leader	+12.567
Pit Indicator	Shows when a driver is in the pits	PIT / OUT
Tire Compound	Current tire compound and age	M (Lap 12)
Position Change	Arrow showing gained/lost positions	Up/Down arrow

Timing Tower Modes

The timing tower can display different information depending on the session and broadcast director's choice:

- Interval mode: Shows gap to car ahead (most common during races)
- Leader mode: Shows gap to the race leader for all drivers
- Practice/Qualifying: Shows best lap times and sector splits
- Pit stop mode: Highlights pit stop duration and positions gained/lost

Timing Tower (Example)

1	VER LEADER	1:31.245
2	NOR +1.234	1:31.489
3	LEC +3.567	1:31.812
4	HAM +5.891	1:32.001
5	PIA +7.123	1:32.156

Example timing tower display

Sector Colors

Each circuit is divided into three sectors (S1, S2, S3). As a driver completes each sector, the time is color-coded to show performance.



Purple - Session Fastest



Green - Personal Best



Yellow - Not Improved

Color	Meaning	Detail
Purple	Overall fastest	Fastest sector time by ANY driver in the session
Green	Personal best	Driver's own best sector time in the session
Yellow	Not a personal best	Slower than the driver's previous best
White	First timed lap	No comparison available yet

Reading Sectors During Qualifying

In qualifying, watch the sector colors to predict lap times before the driver crosses the line. Three purple sectors = potential pole lap. A yellow sector means the driver has lost time and the lap may not improve their position.

Mini Sectors

Each sector is further divided into mini sectors (approximately every 200-300 meters). These provide even more granular speed comparisons and are shown on detailed telemetry overlays.

Tire Information

Tire graphics are critical for understanding race strategy. Pirelli supplies all tires with standardized color coding.

Dry Compounds



Red - Soft



Yellow - Medium

White - Hard

Compound	Color	Range	Characteristics
Hard	White	C1-C2	Most durable, least grip, longest stints
Medium	Yellow	C2-C4	Balanced durability and performance
Soft	Red	C3-C5	Maximum grip, fastest degradation, qualifying

Wet Compounds



Green - Intermediate



Blue - Full Wet

Compound	Color	Usage
Intermediate	Green	Damp/drying track; light rain or drying conditions
Full Wet	Blue	Heavy rain; standing water on track

On-Screen Tire Information

- Tire compound icon (colored circle) next to driver name on timing tower
- Tire age in laps (e.g., 'Lap 15' means 15 laps since fitting)
- Tire performance curves shown during strategy analysis graphics
- Pit stop window predictions overlaid on race progression charts
- Used vs. new tire indicator (some broadcasts show this distinction)

Compound Selection

Pirelli selects three of the five dry compounds (C1-C5) for each Grand Prix. These are labeled Hard, Medium, and Soft regardless of which C-numbers they actually are. A C3 tire might be the 'Hard' at one race and the 'Soft' at another.

Telemetry & Data Overlays

Modern F1 broadcasts include detailed telemetry graphics that show real-time car data. These are especially prominent during onboard camera views.

Speed & Gear

- Speed (km/h or mph): Displayed as a large number on onboard views
- Gear indicator: Shows current gear (1-8), often with RPM bar
- Speed trace: Line graph comparing speeds of two drivers through a lap
- Speed trap: Maximum speed recorded at a specific point on the straight

Throttle & Brake

- Throttle application: Bar showing 0-100% throttle input
- Brake pressure: Bar showing braking force (often red-colored)
- Combined view shows both inputs simultaneously during onboard laps
- Useful for comparing driving styles into braking zones

ERS & Battery (New for 2026)

With the dramatically increased electric power in 2026, battery and ERS graphics become much more important:



- Battery state of charge: Percentage or bar showing remaining energy
- ERS deployment: Visual indicator of when electric power is being used
- ERS harvesting: Shows when the MGU-K is recovering energy under braking
- X-Mode indicator: Shows when active aero is in low-drag configuration
- Power split graphic: Pie chart or bar showing ICE vs. electric power
- Energy management is now a key strategic element visible on broadcasts

Why ERS Matters More in 2026

With ~50% of power from electric sources, battery management becomes as important as tire strategy. Drivers who manage energy better on high-demand circuits will have a significant advantage. Watch the battery graphic to see who is deploying and harvesting efficiently.

Flags & Their Meanings

Flags are the primary means of communication between race control and drivers. They are shown on-screen and at marshalling posts around the circuit.

Flag	Color/Pattern	Meaning
------	---------------	---------

Flag	Color/Pattern	Meaning
Green	Solid green	Track clear, racing conditions, session start
Yellow	Solid yellow	Danger ahead - slow down, no overtaking in zone
Double Yellow	Two yellow flags	Great danger - be prepared to stop
Red	Solid red	Session stopped - return to pit lane immediately
Blue	Solid blue	Let faster car through (lapped traffic)
White	Solid white	Slow-moving vehicle ahead (recovery, medical)
Black	Solid black	Driver disqualified (shown with car number)
Black/Orange	Black + orange circle	Car has mechanical issue - must pit
Black/White	Diagonal halves	Unsportsmanlike behavior warning
Yellow/Red	Red + yellow stripes	Slippery surface (oil, water, debris)
Checkered	Black/white checks	Session or race finished

Safety Car & VSC Graphics

- SC (Safety Car): Yellow 'SC' banner across timing tower; field bunches up
- VSC (Virtual Safety Car): 'VSC' indicator; cars maintain delta time
- SC ending: 'SC IN THIS LAP' message before green flag restart
- Red flag: Red banner replaces all other graphics

Weather Graphics

Weather information is crucial in F1, as conditions directly affect tire choice, strategy, and car setup.

On-Screen Weather Data

Metric	What It Shows	Why It Matters
Air Temp	Ambient air temperature (C/F)	Affects engine cooling and aero
Track Temp	Road surface temperature	Major impact on tire grip/degradation
Humidity	Moisture in the air (%)	Affects engine performance
Wind Speed	Wind velocity and direction	Impacts aero balance, especially on straights
Rain Probability	Chance of rain (%)	Triggers strategy decisions
Radar	Rain radar overlay of circuit	Shows approaching weather systems

Weather Strategy

A rain probability increase during a race creates massive strategic decisions: pit now for intermediates or gamble on staying out? Watch the radar graphic to see if rain is actually hitting the circuit. Different sectors can have different conditions simultaneously.

Additional Broadcast Elements

Battle Graphics

- Head-to-head comparison between two close drivers
- Gap timer showing live interval between battling cars
- Overtake probability (experimental AI-driven graphic)
- Historical comparison of laps between two drivers

Strategy Graphics

- Pit stop window: Predicted optimal pit window for each driver
- Undercut/overcut analysis: Shows if an early or late stop is beneficial
- Tire life remaining: Estimated laps left on current compound
- Race progression chart: Lap-by-lap position chart for all drivers

Driver Communications

- Team radio transcripts displayed on screen
- Driver-of-the-Day voting graphics
- Penalty notifications with explanation
- Track limit warnings and violations

Quick Reference Card

Print this page as a single-sheet cheat sheet for race day.

Sector Colors at a Glance



Purple = Session Fastest



Green = Personal Best



Yellow = Not Improved

Tire Colors at a Glance

 Soft (Red)

 Medium (Yellow)

 Hard (White)

 Inter (Green)

 Wet (Blue)

Common Flag Summary

Green	Yellow	Red	Blue	Checkered
Clear	Danger	Stopped	Move over	Finished

Points (Race / Sprint)

1st	2nd	3rd	4th	5th	6th	7th	8th	9th	10th
25/8	18/7	15/6	12/5	10/4	8/3	6/2	4/1	2/-	1/-

+ 1 point for fastest lap in the Grand Prix (if finishing in top 10).